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Digital Contracts: Simple Tools for Pricing Complex Derivatives*

This article presents a simple, unified approach for valuing a variety of financial assets using digital contracts. Three types of digitals are used: a digital option paying either one dollar or nothing, a digital share paying nothing or converting into one share of the underlying asset; and a first-touch digital paying one dollar the first time that the price of the underlying stock moves into some specified region. It is shown how the values of these three types of digitals can be determined for a wide variety of payoff events and how they can be combined to price complex contracts.

I. Introduction

The number and variety of derivative contracts available over the counter, on exchanges, and through private placements has grown substantially in the past decades. In fact, there is a small cottage industry devoted simply to the classification of contracts into hierarchies. Meanwhile, financial academicians and practitioners are busily employed deriving formulas to value these increasingly complex contracts.

Unfortunately, most of these formulas are quite specialized. In general, each can be applied only to the specfic asset for which it was derived or to other closely related contracts, and often it is not at all obvious how one formula can be applied even for very similar contracts. This article presents a simple and unified approach for determining formulas for a wide variety of financial assets that use digital contracts as simple building blocks. Digital contracts are the simplest building blocks because their payoffs are either ''on'' or ''off.'' We will use three types of digital contracts. A digital option pays at maturity either one dollar (on) or nothing (off) depending on its payoff event. Similarly, at maturity a digital share pays nothing (off) or converts into one share of the underlying basis asset (on). A first-touch digital pays one dollar the first time that the price of the underlying stock moves into some specified

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region (on); if this does not happen by the maturity date, the contract expires and is worthless (off).

The valuation procedure with these digitals has three steps. The (risk-neutral) probability of a particular payoff event is determined by standard statistical methods. Armed with this, the formulas for the digital contracts are constructed. Then, these instruments are used to value financial derivative contracts. Even in those cases in which no formula can be derived, the methods developed here can be used to determine simple approximations of arbitrary accuracy.

The outline of the article is as follows. Section II of the article reviews the basics of derivatives pricing. Section III introduces the simplified digital approach. Sections IV–VII apply this method to specific examples. Section VIII shows how to compute delta and other Greek measures with digitals. Section IX concludes the article with a discussion of extensions.

II. Valuing European Derivative Securities with Digital Contracts

A pure European-style option is one with a single payoff received on a maturity date known at the contract's inception. The level of the payoff may, of course, depend on events that occur between the issuance date and the maturity date.

The payoffs on most European contracts are piecewise linear in the stock price (or other underlying) on the maturity date. The value of any such contract can be represented as

$$\sum_i a_i \mathcal{D}(S, t; T; \mathcal{E}_i) + \sum_j b_j \mathcal{S}(S, t; T; \mathcal{E}_j). \tag{1}$$

Here $\mathcal{D}(S, t; T; \mathcal{E})$ is the value at time t of receiving \$1 at time T, the maturity date, if and only if the event $\mathcal{E}$ occurs; $\mathcal{S}(S, t; T; \mathcal{E})$ is the value at time t of receiving one share of the stock at time T (excluding any intervening dividends), if and only if the event $\mathcal{E}$ occurs. In general, the probability of $\mathcal{E}$ depends on the current stock price, so the values of $\mathcal{D}$ and $\mathcal{S}$ also depend on S.

Two simple examples illustrate this digital portfolio pricing approach. A standard European put option can be represented as $X\mathcal{D}(S, t; T; S_T < X) - \mathcal{S}(S, t; T; S_T < X)$. That is, X dollars are received and one share of stock is given up when the stock price at maturity is below the strike price so that the option expires in-the-money. A down-and-out call option can be represented as $\mathcal{S}(S, t; T; S_T > X \cap S_{\min(0,T)} > K) - X\mathcal{D}(S, t; T; S_T > X \cap S_{\min(0,T)} > K)$. For this option, X dollars are paid to receive a share of stock if the option is in-the-money at maturity and the stock price never fell below the knock-out price (out-strike) K. Many other examples can be constructed similarly.

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A contract such as $\mathcal{D}(\cdot)$, which pays a fixed amount if and only if some event occurs and pays nothing, otherwise has been called a binary option, or a cash-or-nothing option.[1] A digital option is a risky type of zero-coupon bond with either full payment or complete ''default,'' as determined by the event $\mathcal{E}$. A contract such as $\mathcal{S}(\cdot)$, which converts to a share of stock if some event occurs, has also been called an ''all-or-nothing'' share. We will use the term ''digitial share.''

Digital options and shares can be valued like any other derivative asset by the Cox-Ross (1976) risk-neutral technique. If the stock price follows a diffusion process with a constant volatility, σ, and has a constant divident yield, q,

$$dS = (\mu - q)Sdt + \sigma Sd\omega, \tag{2}$$

then the value of any claim with a path-independent payoff event, $\mathcal{E}$, satisfies the Black-Scholes (1973) partial differential equation

$$\frac{1}{2}\sigma^2 S^2 F_{SS} + (r - q)SF_S + F_t - rF = 0. \tag{3}$$

The maturity conditions for the digitals are[2]

$$\begin{aligned} \mathcal{D}(S, t; T; \mathcal{E}) &= \begin{cases} 0 & \text{if } \mathcal{E} \text{ has not occurred} \\ 1 & \text{if } \mathcal{E} \text{ has occurred} \end{cases} \\ \mathcal{S}(S, t; T; \mathcal{E}) &= \begin{cases} 0 & \text{if } \mathcal{E} \text{ has not occurred} \\ S & \text{if } \mathcal{E} \text{ has occurred.} \end{cases} \end{aligned} \tag{4}$$

From the Black-Scholes (1973) model, we know that the values of a digital option and a digital share for the event $\mathcal{E} = \{S_T > X\}$ are

$$\begin{aligned} \mathcal{D}(S, t; T; S_T > X) &= e^{-r(T-t)}\Phi(h^-), \\ \mathcal{S}(S, t; T; S_T > X) &= Se^{-q(T-t)}\Phi(h^+), \end{aligned} \tag{5}$$

1. The terms ''binary'' and ''digital option'' are generic and can be applied to contracts whose payoff events are complex. For example, a digital option might pay \$1 if the stock price at maturity is between \$40 and \$60 or if the minimum stock price over the option's life is less than \$30. The name ''cash-or-nothing'' is generally applied only to options with the simple event that the stock price at maturity is greater or less than a certain level. In particular, a cash-or-nothing call with a strike price of \$50 pays \$1 if the stock price at maturity is greater than \$50. A cash-or-nothing put pays \$1 if the stock price at maturity is less than \$50.

2. Two boundary conditions in S are also needed. These will depend on the specific contract. Generally, they will set the value of the contract to zero (or otherwise bound its value) at $S = 0$ or $S = \infty$.

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where

$$h^{\pm} \equiv \frac{\ln(S/X) + \left(r - q \pm \frac{1}{2}\sigma^2\right)(T - t)}{\sigma\sqrt{T - t}}.$$

The digital option's value is equal to the risk-neutral probability of the payoff event multiplied by the risk-free discount factor. The digital share's value is the stock price, excluding the present value of the dividends multiplied by the risk-neutral probability of the payoff event adjusted for the increased expected value of the stock price conditional on the event $S_T > X$.

The pricing of other European derivatives with piecewise linear and path-independent payoffs only requires valuing digital options and shares for events of the type $L < S_T < H$. Any other path-independent event can be described as the union of such events.[3] These digitals can be computed as the difference between two unlimited range digitals given in

$$\begin{aligned}\mathcal{D}(S, t; T; L < S_T < H) = \mathcal{D}(S, t; T; L < S_T) \\ - \mathcal{D}(S, t; T; H \leq S_T),\end{aligned} \tag{6}$$

and similarly for digital shares.[4]

III. Digital Values for Other Single State Variable Events

For path-dependent events the valuation of digital options is still relatively easy. Since the payoff on a digital option is always one or zero, the value of a digital option is simply the risk-neutral probability of the payoff event $\mathcal{E}$ multiplied by the default-free discount factor,

$$\mathcal{D}(S, t; T; \mathcal{E}) = e^{-r(T-t)}\text{Prob}\{\mathcal{E}|S_t = S\}, \tag{7}$$

just as with path-independent payoffs. The probabilities for many such events are known or can be determined from the theory of diffusion processes.

3. In the Black-Scholes framework (1973) S_T is a continuous variable, and one could imagine a contract with a payoff of \$1 if S_T is rational and zero if S_T is irrational. An event of this type cannot be represented as a union of range events, $L < S_T < H$. Such exotic contracts are of little practical interest and could not be implemented when S_T is discrete in any case.

4. For the Black-Scholes model (1973) or any diffusion model without singular dynamics, we may use $\mathcal{D}(S, t; T; S_T > L) - \mathcal{D}(S, t; T; S_T > H)$ since the events $S_T > H$ and $S_T \geq H$ have the same probability and, therefore, the same digital values. For ranges with a lower limit of zero, we use $\mathcal{D}(S, t; T; S_T > 0) = e^{-r(T-t)}$ and $\mathcal{S}(S, t; T; S_T > 0) = Se^{-q(T-t)}$.

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The usual approach to valuing a digital share would be to integrate its payoff, S_T, with respect to the risk-neutral probability distribution to determine the risk-neutral expectation and then to discount. However, the value of a digital share can also be determined directly from the formula for the corresponding digital option without integration, as I now show.

Let $G(S, t) \equiv \mathcal{S}(S, t; T; \mathcal{E})/Se^{(r-q)(T-t)}$. Then the partial derivatives of $\mathcal{S}$ are

$$\begin{aligned} \frac{\partial \mathcal{S}}{\partial S} &= e^{(r-q)(T-t)}(G + SG_S), \\ \frac{\partial^2 \mathcal{S}}{\partial S^2} &= e^{(r-q)(T-t)}(2G_S + SG_{SS}), \\ \frac{\partial \mathcal{S}}{\partial t} &= e^{(r-q)(T-t)}[SG_t - (r - q)SG]. \end{aligned} \tag{8}$$

Substituting these derivatives into the valuation partial differential equation (3) gives

$$Se^{(r-q)(T-t)}\left[\frac{1}{2}\sigma^2 S^2 G_{SS} + (r - q + \sigma^2)SG_S + G_t - rG\right] = 0. \tag{9}$$

The maturity condition for the digital share is $\mathcal{S}(S, T) = S$ if $\mathcal{E}$ has occurred and zero otherwise; therefore, the maturity condition for G is $G(S, T) = \mathcal{S}(S, T)/S = 1$ if $\mathcal{E}$ has occurred and zero otherwise. Clearly, G has the same maturity condition as a digital option and satisfies the same partial differential equation, with q being replaced by $q - \sigma^2$. Thus, the value of a digital share for any payoff event $\mathcal{E}$ is equal to $Se^{(r-q)(T-t)}$ multiplied by the value of the digital option for the same event with a modified dividend yield, $q - \sigma^2$, which is substituted for the original dividend yield, q,[5]

$$\mathcal{S}(S, t; T, r, q, \sigma; \mathcal{E}) = Se^{(r-q)(T-t)}\,\mathcal{D}(S, t; T, r, q - \sigma^2, \sigma; \mathcal{E}). \tag{10}$$

This result can be verified for the Black-Scholes digitals given in (5) and other known cases. For example, a down-and-out call has a

5. The relation in (10) is the result of a change of measure much like the change of measure from the original to risk-neutral probabilities in the Cox-Ross technique (1976). In each case the drift of the stock price has been altered. This can also be expressed as a change in the interest rate as in Cox-Ross; however, that choice is not as convenient for multiasset contracts when different adjustments must be made for the digital share on each. Alternatively, the result can be written as $\mathcal{S}(S, t; T; \mathcal{E}) = Se^{(r-q)(T-t)}\,\mathcal{D}(Se^{\sigma^2(T-t)}, t; T; \mathcal{E})$. This representation is convenient when the asset does not have continuously paid dividends. It should be noted that this is a formal change made in the formula and not an introduction of an asset with the price $Se^{\sigma^2(T-t)}$. For example, a barrier is breached when S moves past the barrier, not when $Se^{\sigma^2(T-t)}$ does.

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payoff event in which the stock price at maturity exceeds the strike price, X, and the minimum stock price over the interval (t, T) exceeds the knock-out barrier, K. From Merton's (1973) valuation of the down-and-out call, we can write the value of the associated digitals as

$$\mathcal{D}(S, t; T; S_T > X \cap S_{\min(t,T)} > K) = e^{-r(T-t)}\left[\Phi(h^-) - \left(\frac{K}{S}\right)^{\gamma-1}\Phi(\zeta^-)\right] \quad (11a)$$

$$\mathcal{S}(S, t; T; S_T > X \cap S_{\min(t,T)} > K) = Se^{-q(T-t)}\left[\Phi(h^+) - \left(\frac{K}{S}\right)^{\gamma+1}\Phi(\zeta^+)\right], \quad (11b)$$

where

$$\gamma \equiv \frac{2(r-q)}{\sigma^2} \quad \zeta^{\pm} \equiv \frac{\ln(K^2/SX) + \left(r - q \pm \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}.$$

It can be easily verified that the relation in (10) is satisfied for these two digitals since $h^- \to h^+$, $\zeta^- \to \zeta^+$ and $\gamma - 1 \to \gamma + 1$ when q is replaced by $q - \sigma^2$.

The rationale for this tranformation and a intuitive explanation for the result can be found in a simple change of numeraire to the stock price. Consider the digital share contract that converts to one share when $S_T > X$. Using a share of stock rather than dollars as the numeraire, this payoff is one (share of stock) whenever X unit (dollar) face-value bonds (worth X/S_T shares) are worth less than one share. This is a zero-one payoff, just like the digital option in the cash numeraire, so its value is the risk-neutral probability of the event, discounted to the present at the risk-free rate. However, since we are pricing in the stock numeraire, we must compute the risk-neutral probability and discount relative to this numeraire. In the stock numeraire, the stock is the risk-free asset since its price is constant. The interest rate is the yield on the risk-free asset—in this case q; so q is both the discount rate and the risk-neutral expected rate of return.[6] The present value of this contract in the stock numeraire is

$$\text{PV}[1 \text{ when } X/S_T < 1] = e^{-q(T-t)}\,\widehat{\text{Prob}}\{X/S_T < 1\}. \quad (12)$$

6. The dividend yield, q, is a pure number since it is the ratio of the dollar value of the dividend, qS, to the dollar value of the share, S. In the stock numeraire it is the same ratio, in this case q to 1, so it is not altered in the change of numeraire, which affects only prices. We will use a hat to denote risk-neutral probabilities and expectations in the stock numeraire.

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The price of the bond in the stock numeraire is just the reciprocal of the stock price in the dollar numeraire, so it also has a lognormal distribution. The variance of $\ln(X/S_T) - \ln(X/S_T)$ is $\sigma^2(T - t)$. The (stock numeraire) risk-neutral expected growth rate of the bond price is the risk-free rate, q, less the ''dividend'' yield on its disbursements, r. So $\hat{\mathrm{E}}[X/S_T] = Xe^{(r-q)(T-t)}/S$ and $\hat{\mathrm{E}}[\ln(X/S_T)]$ is the logarithm of this, less half the logarithmic variance as usual:

$$\hat{\mathrm{E}}[\ln(X/S_T)] = \ln(X/S_t) + \left(q - r - \frac{1}{2}\sigma^2\right)(T - t). \tag{13}$$

Therefore, the risk-neutral probability in the stock numeraire is

$$\begin{aligned}\widehat{\mathrm{Prob}}\{X/S_T < 1\} &= \Phi\left[\frac{\ln(1) - \ln(X/S) - \left(q - r - \frac{1}{2}\sigma^2\right)(T - t)}{\sigma\sqrt{T - t}}\right] \\ &= \Phi\left[\frac{\ln(S/X) + \left(r - q + \frac{1}{2}\sigma^2\right)(T - t)}{\sigma\sqrt{T - t}}\right] = \Phi(h^+).\end{aligned} \tag{14}$$

The present value of this payment is this probability discounted at an ''interest rate'' of q or $e^{-q(T-t)}\Phi(h^+)$ shares. Since each share is currently worth S dollars, this value is $Se^{-q(T-t)} \cdot \Phi(h^+)$ dollars.

The importance of this valuation technique is that we can easily price any new option or derivative contract for which the probability distribution of the payoff event is known without having to perform an integration. The next four sections examine some special cases with examples.

IV. Options with Nonlinear Payoffs

Recently, options have been introduced with nonlinear payoffs. The most common type of nonlinear option is the power option or power cap. Power options have a payoff based on some power of the basis, usually the second power. Power options are most often used with fixed income derivatives, such as the London Interbank Offer Rate squared note, but they have also begun trading on foreign currencies and equities.

To value options with nonlinear payoffs, we can adapt the basic digital method. Let $\mathcal{P}(S;\ t;\ T;\ \alpha;\ \mathcal{E})$ represent a power digital that is the value of a payoff of S^α at time T if the event $\mathcal{E}$ has occurred. To value

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the power digital, we introduce the artificial asset $Z \equiv S^{\alpha}$. By Itô's lemma the risk-neutral evolution of Z is

$$\begin{aligned} dZ &= \alpha S^{\alpha-1}[(r - q)S dt + \sigma S d\omega] + \frac{1}{2}\alpha(\alpha - 1)S^{\alpha-2}\sigma^2 S^2 dt \\ &= (r - q_{\alpha})Z dt + \alpha\sigma Z d\omega, \end{aligned} \tag{15}$$

where

$$q_{\alpha} \equiv q + (1 - \alpha)\left(r - q + \frac{1}{2}\alpha\sigma^2\right).$$

The term q_{α}, which is the difference between the expected growth rate and the interest rate, can be treated as if it were a continuously paid dividend. The ''dividend'' may be negative for $\alpha > 1$, but this is irrelevant for the mathematics of the pricing. Since a power digital pays one unit of $Z = S^{\alpha}$, the value of a power digital can be written immediately as.

$$\mathcal{P}(S, t; T; \alpha; r, q, \sigma; \mathcal{E}) = \mathcal{S}(S^{\alpha}; t; T; r, q_{\alpha}, \alpha\sigma; \mathcal{E}). \tag{16}$$

For example, an asymmetric power call has a payoff of $\mathrm{Max}(S^{\alpha} - X^{\alpha}, 0)$. The payoff event for this call is $S_T > X$ (for $\alpha > 0$). The power digital for a power-α option is worth

$$\begin{aligned} \mathcal{P}(S, t; T; \alpha; r, q, \sigma; S_T > X) &= \mathcal{S}(S^{\alpha}, t; T; r, q_{\alpha}, \alpha\sigma; S_T^{\alpha} > X^{\alpha}) \\ &= S^{\alpha} e^{-q_{\alpha}(T-t)}\Phi(h_{\alpha}^{+}), \end{aligned} \tag{17}$$

where

$$\begin{aligned} h_{\alpha}^{+} &\equiv \frac{\ln(S^{\alpha}/X^{\alpha}) + \left(r - q_{\alpha} + \frac{1}{2}\alpha^2\sigma^2\right)(T - t)}{\alpha\sigma\sqrt{t - t}} \\ &= \frac{\ln(S/X) + \left(r - q + \alpha - \frac{1}{2}\sigma^2\right)(T - t)}{\sigma\sqrt{T - t}}. \end{aligned}$$

The case $\alpha < 0$ can be handled similarly. In this case the option is in-the-money for $S < X$, so we would use $-h_{\alpha}^{+}$ as the argument to the cumulative normal.

Values for other functional forms of the payoff can be approximated to whatever accuracy is required by linear combinations (splines) of power and regular digitals. In the approximation, tractable payoffs are substituted for the actual payoffs or events.

For example, suppose the payoff on a contract is $H(S_T)$. The payoff

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can be fit approximately by choosing the weights w_{ij} and the knot points X_j so that

$$H(S) \approx \sum_{i=1}^{I_j} w_{ij} S^{\alpha_{ij}} \quad \text{for } X_j < S < X_{j+1} \tag{18}$$

in J distinct regions.[7] Standard spline methods can be used to get the best fit for the payoff, although here splines are not constrained to be polynomial, as is typically the case. Once the fit has been constructed, the contact is valued approximately as

$$\sum_{j=1}^{J} \sum_{i=1}^{I_j} w_{ij} \mathcal{P}(S,\ t;\ T;\ \alpha_{ij};\ X_j < S_T < X_{j+1}). \tag{19}$$

Example: DAX Power Warrant

In May 1995, Trinkaus & Burkhardt issued power warrants that were written on 10-year Bunds, the DAX index, and the dollar-Deutschmark exchange rate. Each of the warrants had a European-style exercise and expired early in 1996. All of the warrants were also capped in total payoff.

The DAX power warrant had a strike price of 2,000 and expired on May 23, 1996. If the DAX was above 2,000 on that date, the payoff would be one ten-thousandth of the square of the difference from 2,000 with a cap of 25; that is, $\text{Min}[0.0001(S_T - 2{,}000)^2, 25]$. If the DAX index was below 2,000, the option would expire worthless. The payoff capped at 25 when the DAX ended above 2,500. If the DAX was in the range 2,000–2,500, the payoff would be

$$\frac{1}{10{,}000}(S_T - 2{,}000)^2 = \frac{1}{10{,}000}(S_T^2 - 4{,}000 S_T + 4{,}000{,}000). \tag{20}$$

Letting $\mathcal{E} \equiv \{2{,}000 < S_T < 2{,}500\}$, the value of the DAX warrant can be expressed as

$$\begin{aligned} \text{DAX warrant} = {} & 25\,\mathcal{D}(S, t; T; S_T > 2{,}500) + 0.0001\,\mathcal{P}(S, t; T; 2; \mathcal{E}) \\ & - 0.4\,\mathcal{S}(S, t; T; \mathcal{E}) + 400\,\mathcal{D}(S, t; T; \mathcal{E}). \end{aligned} \tag{21}$$

V. Digital Values for Multiple State Variable Events

If the contract to be valued depends on more than one state variable, then a similar approach can be employed. Suppose a derivative contract

7. Note that, for $\alpha = 0$ and $\alpha = 1$, the power-digital is identical to a digital option or share, respectively.

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depends on the price of an asset S and another state variable (possibly another asset price) Y. The risk-neutral dynamics for Y are[8]

$$dY = (\mu - \lambda)dt + v d\omega_Y. \tag{22}$$

Here, μ is the true expected change in the state variable Y, and λ is the risk-neutral adjustment to this mean. The instantaneous correlation between changes in S and Y is ρ. The partial differential equation for pricing contracts that depend on S and Y is

$$\begin{aligned} 0 = {} & \frac{1}{2}\sigma^2 S^2 F_{SS} + \rho\sigma S v F_{SY} + \frac{1}{2}v^2 F_{YY} \\ & + (r - q)SF_S + (\mu - \lambda)F_Y - rF + F_t. \end{aligned} \tag{23}$$

As before, the value of a digital option that pays \$1 if and only if $\mathscr{E}$ occurs is

$$\mathscr{D}(S, Y, t; T; \mathscr{E}) = e^{-r(T-t)}\text{Prob}(\mathscr{E}|S_T = S, Y_T = Y). \tag{24}$$

To value the digital share for this event, let $G(S, Y, t) \equiv \mathscr{S}(S, Y, t; T; \mathscr{E})/Se^{(r-q)(T-t)}$, as before. The partial derivatives of $\mathscr{S}$ with respect to S are as given in (8); the remaining partial derivatives are

$$\begin{aligned} \frac{\partial \mathscr{S}}{\partial Y} &= e^{(r-q)(T-t)} S G_Y, \\ \frac{\partial^2 \mathscr{S}}{\partial Y^2} &= e^{(r-q)(T-t)}\, S G_{YY}, \\ \frac{\partial^2 \mathscr{S}}{\partial S \partial Y} &= e^{(r-q)(T-t)}\, (G_Y + S G_{SY}). \end{aligned} \tag{25}$$

Substituting these derivatives into (23) and eliminating the common factor gives

$$\begin{aligned} 0 = {} & \frac{1}{2}\sigma^2 S^2 G_{SS} + \rho\sigma S v G_{SY} + \frac{1}{2}v^2 G_{YY} \\ & + (r - q + \sigma^2)SG_S + (\mu - \lambda + \rho\sigma v)G_Y - rG + G_t. \end{aligned} \tag{26}$$

The maturity condition for the digital share is $\mathscr{S}(S, Y, T) = S$ if $\mathscr{E}$ has occurred and zero otherwise; therefore, the maturity condition for G is $G(S, Y, T) = 1$ if $\mathscr{E}$ has occurred and zero otherwise, as before. Clearly, G has the same maturity condition as the digital option for $\mathscr{E}$

8. See Cox, Ingersoll, and Ross (1985) for a development of the general partial differential pricing equation for multiple state variables.

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and satisfies the same partial differential equation, with q being replaced by $q - \sigma^2$ and with $\mu - \lambda$ being replaced by $\mu - \lambda + \rho\sigma v$. Therefore, as before,[9]

$$\begin{aligned}\mathcal{S}(S, Y, t; T, r, q, \sigma, \mu - \lambda, v, \rho; \mathcal{E}) \\ = Se^{(r-q)(T-t)}\mathcal{D}(S, Y, t; T, r, \\ q - \sigma^2, \sigma, \mu - \lambda + \rho\sigma v, v, \rho; \mathcal{E}).\end{aligned} \tag{27}$$

For more than two state variables, the same general rule applies. To price a digital share, the digital option formula for the same event is used. The drift term of each state variable is altered by the addition of the covariance between the rate of return on the asset to be received and the change in the level of the state variable. So

$$\mathrm{Cov}\left[\frac{\mathrm{d}S}{S}, \mathrm{d}S\right] = \sigma^2 S\mathrm{d}t \tag{28}$$

is added to the drift of S, while

$$\mathrm{Cov}\left[\frac{\mathrm{d}S}{S}, \mathrm{d}Y_i\right] = \rho_i\sigma v_i\mathrm{d}t \tag{29}$$

is added to the drift of each other state variable Y_i.

Rainbow options are an important class of multiple-state variable derivative contracts. Rainbow options have payoffs that depend on the prices of two or more underlying assets. Using the formula above, the digital shares on two assets, which convert into one share of S_1 and one share of S_2 if event $\mathcal{E}$ occurs, are[10]

$$\begin{aligned}&\mathcal{S}_1(S_1, S_2, t; T, r, q_1, \sigma_1, q_2, \sigma_2, \rho; \mathcal{E}) \\ &= S_1e^{(r-q_1)(T-t)}\mathcal{D}(S_1, S_2, t; T, r, q_1 - \sigma_1^2, \sigma_1, q_2 - \rho\sigma_1\sigma_2, \sigma_2, \rho; \mathcal{E});\end{aligned} \tag{30a}$$

$$\begin{aligned}&\mathcal{S}_2(S_1, S_2, t; T, r, q_1, \sigma_1, q_2, \sigma_2, \rho; \mathcal{E}) \\ &= S_2e^{(r-q_2)(T-t)}\mathcal{D}(S_1, S_2, t; T, r, q_1 - \rho\sigma_1\sigma_1, \sigma_1, q_2 - \sigma_2^2, \sigma_2, \rho; \mathcal{E}).\end{aligned} \tag{30b}$$

9. If μ, λ, and $\rho\sigma v$ are constants, then eq. (27) describes the digital share as a function. If these variables are not constants, but $\rho\sigma v$ has has the same functional form in S, Y, and t as does $\mu - \lambda$, then the results still hold with the obvious interpretation of the substitution for $\mu - \lambda$. If the functional forms are different, then the substitution technique is still valid, but the altered partial differential eq. (26) may have a different solution from the original partial differential eq. (23).

10. As with single state variable digitals, this result can also be written as $\mathcal{S}_i(S_1, S_2, t; T; \mathcal{E}) = S_ie^{(r-q_i)(T-t)}\,\mathcal{D}(S_1e^{\rho_{i1}\sigma_i\sigma_1(T-t)}, S_2e^{\rho_{i2}\sigma_i\sigma_2(T-t)}; t; T; \mathcal{E})$, with the same caution about interpretation as given in n. 5 above.

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For example, the event $\{S_{1T} > X_1 \cap S_{2T} > X_2\}$ has a risk-neutral probability of

$$\text{Prob}\{S_{1T} > X_1 \text{ and } S_{2T} > X_2\} = \Phi_2(h_1^-, h_2^-, \rho), \tag{31}$$

where

$$h_i^- \equiv \frac{\ln(S_i/X_i) + \left(r - q_i - \frac{1}{2}\sigma_i^2\right)(T - t)}{\sigma_i\sqrt{T - t}}$$

and where $\Phi_2(\cdot)$ is the standard bivariate cumulative normal distribution function. The values of the digital option and shares for this event are

$$\begin{aligned} \mathcal{D}(S_1, S_2, t; T; S_{1T} > X_1 \cap S_{2T} > X_2) &= e^{-r(T-t)}\Phi_2(h_1^-, h_2^-, \rho), \\ \mathcal{S}_i(S_1, S_2, t; T; S_{1T} > X_1 \cap S_{2T} > X_2) &= S_i e^{-q_i(T-t)}\Phi_2(h_{1i}^+, h_{2i}^+, \rho), \end{aligned} \tag{32}$$

where

$$h_{ii}^+ \equiv h_i^- + \sigma_i\sqrt{T - t},\ h_{ij}^+ \equiv h_i^- + \rho\sigma_j\sqrt{T - t}.$$

Note the asymmetry in h_{ii}^+ and h_{ij}^+ for the digital share because h_{ii}^+ is adjusted with the variance and h_{ij}^+ is adjusted with the covariance.

Relative price restrictions are also required for pricing some rainbow contracts. For a better-of-the-two option, for example, the payoff events include outcomes such as $S_{1T} > S_{2T}$ or, equivalently, $S_{1T}/S_{2T} > 1$. Since the ratio S_{1T}/S_{2T} has a lognormal distribution, the probabilities of such events can be determined by analogy with (31). For example, the probability for the event $\{S_{1T} > X \cap S_{1T} > \theta S_{2T}\}$ can be determined as

$$\text{Prob}\{S_{1T} > X \cap S_{1T}/S_{2T} > \theta\}. \tag{33}$$

The logarithm of the ratio S_1/S_2 is normally distributed with a risk-neutral expected change of $q_2 - q_1 + 1/2(\sigma_2^2 - \sigma_1^2)$, a variance of $\Upsilon^2 \equiv \sigma_1^2 - 2\rho\sigma_1\sigma_2 + \sigma_2^2$, and a covariance with changes in S_1 of $\sigma_1^2 - \rho\sigma_1\sigma_2$. Using (31),

$$\text{Prob}\{S_{1T} > X_1 \cap S_{1T} > \theta S_{2T}\} = \Phi_2(h_1^-, H^-, (\sigma_1 - \rho\sigma_2)/\Upsilon), \tag{34}$$

where

$$H^- \equiv \frac{\ln(S_1/\theta S_2) + \left(q_2 + \frac{1}{2}\sigma_2^2 - q_1 - \frac{1}{2}\sigma_1^2\right)(T - t)}{\Upsilon\sqrt{T - t}}.$$

To value the digital share $\mathcal{S}_1$ for this event, we replace q_1 by $q_1 - \sigma_1^2$ and q_2 by $q_2 - \rho\sigma_1\sigma_2$. To value the digital share $\mathcal{S}_2$ for this

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event, we replace q_1 by $q_1 - \rho\sigma_1\sigma_2$ and q_2 by $q_2 - \sigma_2^2$. So the values of the digital option and shares for this event are

$$\begin{aligned}
&\mathcal{D}(S_1, S_2, t; T; S_{1T} > X_1 \cap S_{1T} > \theta S_{2T}) = \\
&\qquad\qquad e^{-r(T-t)}\Phi_2(h_1^-, H^-, (\sigma_1 - \rho\sigma_2)/\Upsilon), \\
&\mathcal{S}_i(S_1, S_2, t; T; S_{1T} > X_1 \cap S_{1T} > \theta S_{2T}) = \\
&\qquad\qquad S_i e^{-q_i(T-t)}\Phi_2(h_{1i}^+, H_i^+, (\sigma_1 - \rho\sigma_2)/\Upsilon),
\end{aligned} \tag{35}$$

where

$$H_1^+ \equiv \frac{\ln(S_1/\theta S_2) + \left(q_2 - q_1 + \frac{1}{2}\Upsilon^2\right)(T - t)}{\Upsilon\sqrt{T - t}},$$

$$H_2^+ \equiv \frac{\ln(S_1/\theta S_2) + \left(q_2 - q_1 - \frac{1}{2}\Upsilon^2\right)(T - t)}{\Upsilon\sqrt{T - t}}.$$

Example: Alternative Options

An alternative option confers the right to buy either n_1 shares of S_1 or n_2 shares of S_2 upon payment of the aggregate strike price X. The payoff is $\mathrm{Max}(n_1S_{1T} - X, n_2S_{2T} - X, 0)$. Its present value, expressed in terms of digitals, is

$$\begin{aligned}
&n_1\mathcal{S}_1(S_1, S_2, t; T; n_1S_{1T} > n_2S_{2T} \cap S_{1T} > X/n_1) \\
&- X\mathcal{D}(S_1, S_2, t; T; n_1S_{1T} > n_2S_{2T} \cap S_{1T} > X/n_1) \\
&+ n_2\mathcal{S}_2(S_1, S_2, t; T; n_2S_{2T} > n_1S_{1T} \cap S_{2T} > X/n_2) \\
&- X\mathcal{D}(S_1, S_2, t; T; n_2S_{2T} > n_1S_{1T} \cap S_{2T} > X/n_2).
\end{aligned} \tag{36}$$

Using the techiques developed in the previous section, this value is

$$\begin{aligned}
&n_1S_1e^{-q_i(T-t)}\Phi_2[h_{11}^+, H_1^+, (\sigma_1 - \rho\sigma_2)/\Upsilon] \\
&- Xe^{-r(T-t)}\Phi_2[h_1^-, H^-, (\sigma_1 - \rho\sigma_2)/\Upsilon] \\
&+ n_2S_2e^{-q_2(T-t)}\Phi_2[h_{22}^+, -H_2^+, (\sigma_2 - \rho\sigma_1)/\Upsilon] \\
&- Xe^{-r\tau}\Phi_2[h_2^-, -H^-, (\sigma_2 - \rho\sigma_1)/\Upsilon],
\end{aligned} \tag{37}$$

where

$$X_1 \equiv X/n_1, \quad X_2 \equiv X/n_2, \quad \theta \equiv n_2/n_1.$$

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VI. Multiple-Date Digitals

To value compound or installment options, we require the risk-neutral probabilities of joint events that are based on the price of a single asset at two or more points of time. One simple way to price contracts of this type is by introducing an artificial second asset and using rainbow digitals.

Denote by T° the relevant early date—the first exercise date of a compound option, for example. Let the ''second asset'' by a portfolio holding one share of the underlying asset until time T°. At time T°, the money is transferred to a risk-free asset paying interest and held until maturity. The ''second'' asset's price is $S_{2t} = S_t$ at times before time T° and $S_{2t} = S_{T^\circ}$ at any time after time T°. In particular, at the option's maturity $S_{2t} = S_{T^\circ}$. This asset pays dividends at the rate q prior to T° and at the rate r after T°.

Since there is no uncertainty in the evolution of S_2 over the final portion of its life, the variance of its return and its correlation with asset one are

$$\begin{aligned} \mathrm{Var}[\ln(S_{2T}/S_{2t})] &= \sigma^2(T^\circ - t) \\ \mathrm{Corr}[\ln(S_{2T}/S_{2t}), \ln(S_T/S_t)] &= \frac{\sigma^2(T^\circ - t)}{\sigma\sqrt{T-t}\,\sigma\sqrt{T^\circ - t}} = \sqrt{\frac{T^\circ - t}{T - t}}. \end{aligned} \tag{38}$$

Therefore, to value dual-date digitals, we employ the rainbow digital formulas with

$$\sigma_2 = \sigma\sqrt{\frac{T^\circ - t}{T - t}} \quad \rho = \sqrt{\frac{T^\circ - t}{T - t}} \quad q_2 = q\frac{T^\circ - t}{T - t} + r\frac{T - T^\circ}{T - t}. \tag{39}$$

These are formal values that are substituted directly into the final formulas. They are not the instantaneous values that appear in the partial differential equation.

Example: Call Option with a Gradual Knock-Out Barrier

The increasing popularity of barrier options has caused some concern among both traders and regulators. Since the deltas of knock-out and knock-in options change abruptly when the barrier is hit, they are difficult to hedge near the barrier. Furthermore, if many traders hold options with near-coincident barriers, as is common in the foreign exchange market, a barrier breach caused by a brief movement and, perhaps, a movement that causes disequilibrium, can precipitate a large number of hedging trades. There has been sentiment among market participants to regulate or even ban barrier options.

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A market solution to this problem has been the introduction of options that do not knock out immediately at their barriers. Two such options are those with soft barriers and those with gradual barriers. A soft-barrier option is knocked out fully or in part, depending on how far past the knock-out price the stock price moves. A gradual-barrier option is knocked out fully or in part, depending on how long the stock price is beyond the barrier.

For example, a down-and-out call with a gradual barrier of K has a payoff of

$$\frac{n_K}{N}\mathrm{Max}(S_T - X, 0), \tag{40}$$

where n_K is the number of days during the total monitoring period of N days for which the closing price (or other monitored price) is above the knock-out price. If the stock price is never below the barrier, then the option's payoff is $S - X$, just like a regular option. For each day on which the stock price is below the knock-out price, however, the option payoff is reduced by $1/N$.[11]

To price this option, we first determine the risk-neutral probability of the N events: the stock price at maturity is above X and the stock price on the nth monitoring date, t^n, is above K; that is, $\mathscr{E}_n \equiv \{S_T > X \cap S_{t^n} > K\}$. Since the logarithms of the stock prices on dates T and t^n have a joint normal distribution, the risk-neutral probability of this event[12] is

$$\mathrm{Prob}\{\mathscr{E}_n\} = \Phi_2(h^-, h_n^-, \rho_n\},$$

where

$$h_n^{\pm} \equiv \frac{\ln(S/K) + \left(r - q \pm \frac{1}{2}\sigma^2\right)(t^n - t)}{\sigma\sqrt{t^n - t}},$$
$$\rho_n \equiv \sqrt{(t^n - t)/(T - \mathrm{t})}. \tag{41}$$

11. Linetsky (1996, 1998) provides a much more extensive examination of gradual-barrier (or step-barrier) options. In particular, he gives a solution for continuous monitoring with a payoff $\mathrm{Max}(1 - \tau_K/T, 0) \cdot \mathrm{Max}(S_T - X, 0)$, where τ_K is the amount of time that the stock price spends below the barrier K. He examines the case when the total monitoring period is more than sufficient to knock out the entire payoff requiring a multiplier of $\mathrm{Max}(1 - \tau_K/T, 0)$ rather than $(1 - \tau_K/T)$, the continuous-time equivalent of the case examined here.

12. In (41), $-h_n^-$ is the number of standard deviations by which the risk-neutral expected stock price on t^n exceeds K, while ρ_n is the correlation between the logarithms of the stock prices on dates t^n and T.

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The gradual knock-out option has the same payoff as a portfolio that is long $1/N$ digital shares and short X/N digital options for each event $\mathscr{E}_n$, so we can immediately write its value as

$$\begin{aligned}\text{Gradual Knock-out} &= \sum_{n=1}^{N} [\mathscr{S}(S, t; T; \mathscr{E}_n) - X\mathscr{D}(S, t; T; \mathscr{E}_n)] \\ &= Se^{-q(T-t)}\frac{1}{N}\sum_{n=1}^{N} \Phi_2(h^+, h_n^+, \rho_n) \\ &\quad - Xe^{-r(T-t)}\frac{1}{N}\sum_{n=1}^{N} \Phi(h^-, h_n^-, \rho_n).\end{aligned} \tag{42}$$

This formula can be easily extended to cases in which the knock-out price is different at each monitoring point.

VII. First-Touch Digitals

To price complex derivative contracts, we often need to know the value of a particular payoff that is received the first time the underlying asset achieves a particular level. For example, in the fall of 1991, the Chicago Board of Options Exchange (CBOE) and the American Exchange (AMEX) introduced capped options on various stock indices.[13] A capped option is like a regular put or call, but the maximum payoff is limited by the cap; the contract is immediately exercised when the option is in-the-money by the cap differential. If the cap differential is C, capped puts are exercised immediately when the stock price falls to $X - C$. Upon this forced exercise, the capped put is worth $X - S = C$. To value a capped put, we must therefore determine the present value of receiving C the first time that the stock price hits $X - C$. We will represent this value by a first-touch digital.

A first-touch digital is a digital contract that pays \$1 when and if a specific event occurs. In particular, $\mathscr{T}(S, t; T; K)$ denotes the value of a first-touch digital that pays \$1 the first time the stock price reaches the level K, provided this occurs before the maturity date, T. If the stock price never reaches K before time T, the claim expires worthless.

The usual approach to valuing a first-touch digital would be to use the first passage time density. (See, e.g., Ingersoll 1977.) If $\Psi(\tau; S, K)$

13. The CBOE lists capped calls and puts on the Standard and Poor's 500 and Standard and Poor's 100 (OEX) indices, while capped options on the Major Market (XMI) and Institutional (XII) indices are traded on the AMEX. On the CBOE capped options, capped calls have their caps set at 30 points above the strike price, while capped puts have their caps set 30 points below the strike. On the AMEX's capped options, the cap differential is 20 points.

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is the risk-neutral probability distribution of the first time that the stock price hits K when its starts at S at time t, then the value of a first-touch digital is

$$\mathcal{T}(S,\ t;\ T;\ K) = \int_t^T e^{-r(\tau-t)} d\Psi(\tau;\ S,\ K). \tag{43}$$

The economics of the problem provides us with an easier solution, however. The first-touch digital is closely related to a digital share with a barrier event. We consider first the situation when the stock pays no dividend. In this case, the first-touch digital is identical in value to the fraction $1/K$ of a barrier digital share.

Suppose $S > K$, so the first-touch digital receives a payment of \$1 when the stock price first falls to the level K. Since the stock price then is K, this \$1 payment can be used to purchase exactly $1/K$ shares of stock. At the maturity of the first-touch contract, this purchase will be worth S_T/K. Therefore, by making this purchase, the holder of a first-touch digital ends up either with $1/K$ shares of stock whenever the stock price reaches K sometime during its life or with nothing if the stock price never falls to K. This is identical to the payoff on $1/K$ digital shares, which pay off if the minimum stock price is less than or equal to K. Similar reasoning applies when $S < K$ relates the first-touch digital to a digital share with a maximum price restriction. Therefore, provided the stock pays no dividends,

$$\mathcal{T}(S,\ t;\ T;\ K) = \begin{cases} \dfrac{1}{K}\mathcal{S}(S,\ t;\ T;\ S_{\min} < K) & \text{for } S > K \\[2ex] \dfrac{1}{K}\mathcal{S}(S,\ t;\ T;\ S_{\max} > K) & \text{for } S < K. \end{cases} \tag{44}$$

If the stock does pay a dividend, we cannot use this same approach to value the first-touch digital. It can still be used to purchase $1/K$ shares when the barrier is first reached, but now this will be more valuable at maturity than receiving the $1/K$ shares if there are any intervening dividends. To use this method, we would need to determine the value of the dividends paid during this interval of random duration.

We saw in Section IV that the stock price raised to a power behaves like an asset price with an altered dividend and volatility. By choosing the power correctly, we can make the altered dividend equal to zero. After doing this we can apply (44). To implement this procedure, we define a new variable

$$Z \equiv S^{\beta-b},$$

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where

$$b \equiv \frac{r - q}{\sigma^2} - \frac{1}{2}, \ \beta \equiv \sqrt{b^2 + 2r/\sigma^2}. \tag{45}$$

By Itô's lemma the risk-neutral dynamics of Z are

$$dZ = rZdt + (\beta - b)\sigma Z d\omega. \tag{46}$$

Clearly, Z evolves just like a price[14] with a risk-neutral expected rate of return of r, no dividends, and a volatility of $(\beta - b)\sigma$, so (44) can be used to value first-touch derivatives written on Z. Since $\beta > b$, Z is below (above) $K^{\beta-b}$ whenever S is below (above) K, and since Z hits $K^{\beta-b}$ when S hits K, first-touch digitals written on Z and on S are identical.

Using (44) to evaluate $\mathcal{T}(Z, t; T; K^{\beta-b})$ and substituting for Z gives the value of the first-touch digitals when there are dividends. These are

$$\mathcal{T}(S, t; T; K) = \begin{cases} \left(\frac{K}{S}\right)^{b-\beta} \cdot \Phi(-h_\beta^+) + \left(\frac{K}{S}\right)^{b+\beta} \Phi(-h_\beta^-) & \text{for } S > K \\ \left(\frac{K}{S}\right)^{b-\beta} \cdot \Phi(h_\beta^+) + \left(\frac{K}{S}\right)^{b+\beta} \Phi(h_\beta^-) & \text{for } S < K, \end{cases} \tag{47}$$

where

$$h_\beta^\pm \equiv \frac{\ln(S/K) \pm \beta\sigma^2(T - t)}{\sigma\sqrt{T - t}}$$

and where b and β are defined in (45).

The present value of a changing first-touch payment can be determined using a portfolio of first-touch digitals with different maturities. For example, suppose the payment is c_1 if the stock price first hits K before $T°$ and is c_2 if the first touch occurs between times $T°$ and T. After T, the contract expires. The present value of this payment is

$$(c_1 - c_2) \cdot \mathcal{T}(S, t; T°, K) + c_2 \cdot \mathcal{T}(S, t; T, K). \tag{48}$$

If the stock price hits K before $T°$, both first-touch digitals will pay off for a net payment of c_1. If the stock price hits K after $T°$ but before T, only the second first-touch digital will pay off.

This procedure can be generalized to handle any payment that

14. In fact, Z is a price. It is the value of a contract that pays $S_T^{\beta-b}$ at time T, where T can be any future time. The choice of β and b ensures $Z = S^{\beta-b}$ at all times.

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changes in steps over time. Let c_i be the payment received if the stock price first hits the barrier K between times T°_{i-1} and T°_i. After time $T^\circ_n \equiv T$, the payment drops to zero (so $c_{n+1} \equiv 0$). The value at time t of this first-touch payment is

$$\sum_{i=1}^{n} (c_i - c_{i+1})\mathcal{T}(S, t; T^\circ_i, K). \tag{49}$$

If the knock out occurs between times T°_{i-1} and T°_i, then the first-touch digitals maturing on dates T°_i and after will still be alive, so the payoff is $\sum_{j=i}^{n} c_j - c_{j+1} = c_i$, as required.

The present value of a continuously changing payment, $c(t)$, that is received the first time a barrier K is hit can be determined as in (49). In all terms except the last, the difference $c_i - c_{i-1}$ is replaced by the differential $-c'(t)\mathrm{d}t$, and the sum becomes an integral:

$$c(T) \cdot \mathcal{T}(S, t; T; K) - \int_t^T c'(\tau) \cdot \mathcal{T}(S, t; \tau; K)\mathrm{d}\tau. \tag{50}$$

Example: The Geothermal Accelerated-Maturity Warrant

Geothermal Resources issued a complex warrant in 1984. It expired on November 15, 1991 (time T); however, the expiration date was accelerated if the stock price reached 150% of the strike price. If it did so before November 15, 1987 (time T°), then the warrant expired on November 15, 1987. If the stock price first reached this level after November 15, 1987, then the warrant expired (and had to be exercised) immediately. The strike price on the warrant was \$13.50, so the relevant first-touch barrier was $1.5 \cdot \$13.50 = \$20.25$. The payoff when the warrant first hit \$20.25 between November 15, 1987, and November 15, 1991, was $\$20.25 - \$13.50 = \$6.75$.

The payoff on the Geothermal warrant can be divided into three parts, each of which we can value:

$$\begin{aligned} &S_T - \$13.50 \text{ at } T \quad \text{if } S_T > \$13.50 \text{ and } S_{\max(t,T)} < \$20.25; \\ &S_{T^\circ} - \$13.50 \text{ at } T^\circ \quad \text{if } S_{T^\circ} > \$13.50 \text{ and } S_{\max(t,T^\circ)} \geq \$20.25; \\ &\$6.75 \text{ when } S \text{ hits } \$20.25 \text{ after } T^\circ \quad \text{if } S_{\max(t,T^\circ)} < \$20.25. \end{aligned} \tag{51}$$

The first part can be valued with a simple barrier digital share and option. The second part can also be valued with a simple barrier digital share and an option that expire at time T°. The third part can be valued as a first-touch digital with a changing payment. The payment is zero until November 15, 1987, and is \$6.75 from then until the warrant expires. The value of the Geothermal warrant is therefore

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$$\begin{aligned}
&\mathcal{S}(S,\, t;\, T,\, S_{\max(t,T)} < \$20.25 \cap S_T > \$13.50) \\
&- \$13.50 \cdot \mathcal{D}(S,\, t;\, T,\, S_{\max(t,T)} < \$20.25 \cap S_T > \$13.50) \\
&+ \mathcal{S}(S,\, t;\, T^\circ,\, S_{\max(t,T^\circ)} \geq \$20.25 \cap S_{T^\circ} > \$13.50) \\
&- \$13.50 \cdot \mathcal{D}(S,\, t;\, T^\circ;\, S_{\max(t,T^\circ)} \geq \$20.25 \cap S_{T^\circ} > \$13.50) \\
&+ \$6.75 \cdot [\mathcal{T}(S,\, t;\, T;\, \$20.25) - \mathcal{T}(S,\, t;\, T^\circ;\, \$20.25)].
\end{aligned} \tag{52}$$

Note that the second set of digitals are ordinary digitals rather than dual-date digitals. Even though their payoff events depend on the stock price at time T°, they also mature then, and so this is the stock price at their maturity.

The first digital share and option pair gives the value of exercising the warrant at maturity, when the stock price has not reached \$20.25, to accelerate the maturity date. The second pair of digitals is the value of exercising the warrant on November 15, 1987, if the stock price has risen above \$20.25 prior to that date. Note that these digitals expire after November 15, 1987, so if the stock price rises above \$20.25 for the first time after this date, these digitals have no payoff.

The final line is the value of a forced exercise of the warrant when the stock price reaches \$20.25 for the first tiime after November 15, 1987. The payoff the first time that the stock price hits \$20.25 is zero if the hit occurs before November 15, 1987, time T°. If the stock price first hits \$20.25 after time T°, the payoff is \$6.75. We value this changing barrier payment as the difference between first-touch digitals expiring at times T° and T with payments of $c_1 = c_2 = \$6.75$, as shown in (48).

VIII. Greek Measures with Digitals

Once a derivative contract has been priced with digitals, its delta can be easily determined, since the delta of a portfolio is just the sum of the deltas of the component parts. For example, the delta of the DAX power warrant whose value was given in (21) is

$$\begin{aligned}
\Delta_{\mathrm{DAX}} = 25 \cdot \Delta_{\mathcal{D}(S,t;T;S_T>2{,}500)} + 0.0001 \cdot \Delta_{\mathcal{P}(S;t;T;2;\mathcal{E})} \\
- 0.4 \cdot \Delta_{\mathcal{S}(S,t;T;\mathcal{E})} + 400 \cdot \Delta_{\mathcal{D}(S,t;T;\mathcal{E})}
\end{aligned} \tag{53}$$

where

$$\mathcal{E} \equiv 2{,}000 < S_T < 2{,}500.$$

The deltas of each of the component parts are known or can be easily determined.

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Alternatively, the deltas can be computed accurately by changing the input price slightly. For example,[15]

$$\Delta_{\mathscr{D}(S,t;T;\mathscr{E})} \approx \frac{\mathscr{D}(S+\epsilon,\, t;\, T;\, \mathscr{E}) - \mathscr{D}(S-\epsilon,\, t;\, T;\, \mathscr{E})}{2\epsilon}. \tag{54}$$

This technique can also be applied to the derivative itself.

Other Greek values can be determined in the same way; the gamma or the vega of a derivative contract is the sum of the gammas or vegas of the digitals in its static replicating portfolio.

IX. Extensions and Conclusion

Most of the results of this article hold for other dividend, interest rate, or volatility structures. With minor modifications, the formulas developed here can be applied to handle those problems.

For example, in valuing digital options, a sloped yield curve can usually be accommodated by using the value of a default-free, zero-coupon bond, $B(t, T)$, in place of the discounting term $e^{-r(T-t)}$. Similarly, if the volatility is not constant, then the product $\sigma^2(T-t)$ is replaced by the total logarithmic volatility of the basis asset over the contract's life, $v^2 = \mathrm{Var}[\ln(S_T/S_t)] = \int_t^T \sigma^2(u)\mathrm{d}u$.

To handle other dividend structures, we replace $Se^{-q(T-t)}$ by the stock price less the present value of the dividends. For example, if there are n discrete dividend payments, each with a yield of y, then we replace $Se^{-q(T-t)}$ by $S(1-y)^n$. If the dividend yield is not constant, then we use

$$S \cdot \exp\left(-\int_t^T q(u)\mathrm{d}u\right) \quad \text{or} \quad S\prod_{i=1}^{n}(1-y_i) \tag{55}$$

for continuous or discrete dividends, respectively.

For digital shares we make these same changes. In addition, we have to adjust the dividend yield with the variance. The simple alteration $q \to q - \sigma^2$ is equivalent to $e^{-q(T-t)} \to e^{(\sigma^2-q)T-t)}$, so for discrete dividends and a changing variance we would use

$$\ln[(1-y)^n] \to \ln[(1-y)^n] + \int_t^T \sigma^2(u)\mathrm{d}u. \tag{56}$$

When exact answers cannot be determined, digitals can be used to give very accurate approximations, as shown in Ingersoll (1998*b*). The methods can also be applied to American or Asian options. See Inger-

15. We use a centered difference in (54) because it is more accurate except near inflection points in the value.

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soll (1998*a*), for American option applications of the digital technology. To value Asian options, we would introduce a second asset whose value at maturity was equal to the average price over the option's life and employ rainbow digitals. With minor changes, interest rate contracts can also be priced with digitals.

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